

# **VENDOR PERFORMANCE ANALYSIS**

*An End-to-End Data Analytics Project*

*Data-Driven Analysis for Procurement, Sales, and Inventory  
Optimization*

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**Technologies Used:**

Python • SQL • SQLite • Pandas • Matplotlib

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## Executive Summary

This project presents an end-to-end analysis of vendor performance using retail purchase, sales, inventory, pricing, and invoice data. The objective was to transform raw transactional data into meaningful business insights that support data-driven purchasing and inventory management decisions.

The analysis began with an ETL pipeline developed in Python to ingest and integrate more than **15.6 million records** from six business datasets into a centralized SQLite database. SQL was then used to aggregate vendor-level metrics, including purchases, sales, gross profit, inventory value, and freight costs. The resulting dataset served as the foundation for exploratory data analysis (EDA) using Pandas and visualization libraries.

The analysis identified high-performing vendors, sales concentration, inventory turnover patterns, capital tied up in unsold inventory, and brands with strong profit potential. Based on these findings, the report provides practical recommendations that can help improve purchasing efficiency, optimize inventory levels, and enhance overall vendor performance.

## 1.1 Background

In the retail industry, effective vendor management plays a crucial role in maintaining profitability and ensuring smooth supply chain operations. Vendors directly influence product availability, procurement costs, inventory levels, and overall business performance. Analysing vendor-related data helps organizations identify high-performing vendors, optimize purchasing decisions, and reduce unnecessary inventory costs.

This project focuses on analysing vendor performance using historical purchase, sales, inventory, and pricing data. By integrating multiple datasets into a centralized database, meaningful insights are generated to support business decision-making.

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## 1.2 Objectives

The primary objectives of this project are:

- Build an ETL pipeline to integrate multiple business datasets into a centralized SQLite database.
- Analyse vendor performance using SQL queries.
- Evaluate sales, purchases, profit margins, inventory turnover, and freight costs.
- Perform exploratory data analysis (EDA) using Python and Pandas.
- Generate business insights and recommendations to support data-driven purchasing and inventory management decisions.

## Dataset Description

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The project utilizes six datasets containing purchase, sales, inventory, pricing, and vendor invoice information. These datasets collectively represent different stages of the retail business process and provide a comprehensive view of vendor performance.

| Dataset         | Description                                                                           |
|-----------------|---------------------------------------------------------------------------------------|
| Begin Inventory | Opening inventory available for each product at the beginning of the analysis period. |
| End Inventory   | Closing inventory remaining after sales and purchases.                                |
| Purchases       | Detailed records of product purchases from vendors, including quantities and costs.   |
| Purchase Prices | Historical purchase prices associated with vendors and brands.                        |
| Sales           | Transaction-level sales data used to calculate revenue and profitability.             |
| Vendor Invoice  | Freight charges, invoice amounts, and payment-related information for vendors.        |

### 3.1 Overview

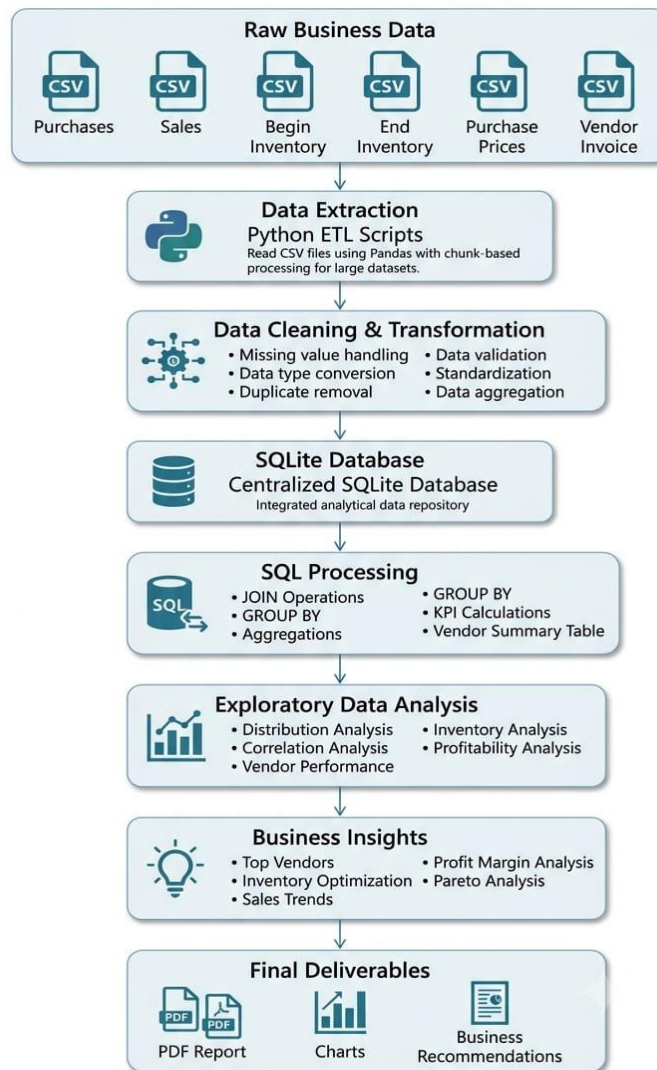
The project follows an Extract, Transform, and Load (ETL) workflow to convert raw retail datasets into a centralized analytical database. The pipeline processes multiple CSV files, performs data cleaning and aggregation, and stores the processed data in SQLite for efficient SQL-based analysis.

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### 3.2 ETL Workflow

#### Vendor Performance Analytics – ETL Workflow

End-to-End Data Processing Pipeline



### 3.3 Extraction

The extraction phase reads multiple retail datasets containing inventory, purchases, sales, pricing, and vendor invoice information. Large datasets were processed efficiently using chunk-based loading in Python to reduce memory consumption while handling millions of records.

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### 3.4 Transformation

The transformation stage focuses on preparing the data for analysis. This includes:

- Cleaning missing and inconsistent values.
  - Standardizing column names and data types.
  - Aggregating vendor-level purchase and sales information.
  - Calculating business metrics such as gross profit, profit margin, inventory value, and freight costs.
  - Creating a consolidated vendor summary table for analysis.
- 

### 3.5 Loading

After transformation, the processed datasets were loaded into a SQLite database. SQL queries were then used to join multiple tables and generate vendor-level performance metrics for exploratory analysis.

### 4.1 Overview

SQL was used to combine data from multiple business tables and calculate key performance indicators (KPIs) for each vendor. Aggregated metrics provided the foundation for all subsequent exploratory data analysis.

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### 4.2 Key Performance Metrics

The following metrics were calculated for every vendor:

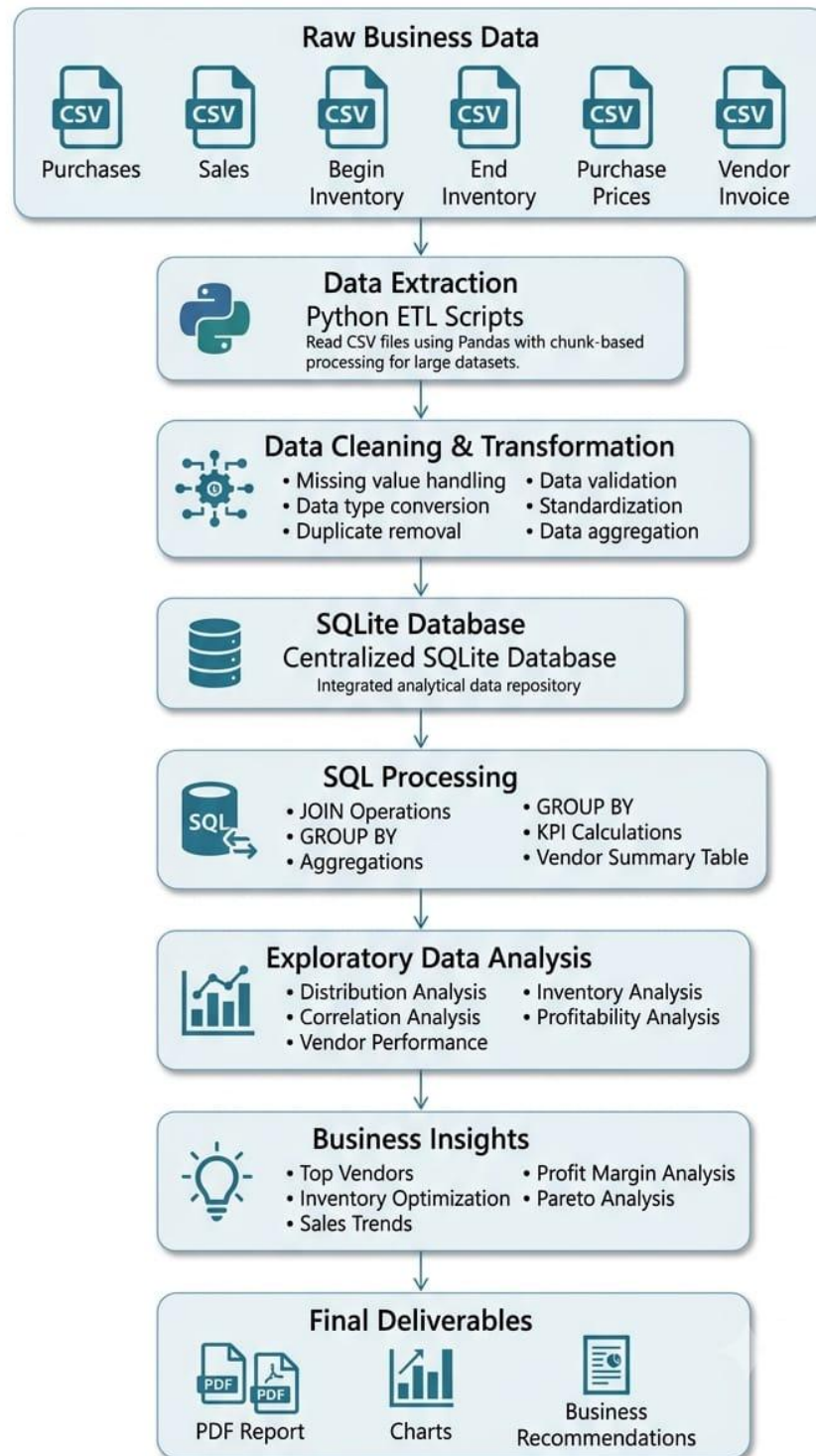
- Total Purchase Cost
- Total Purchase Quantity
- Total Sales Revenue
- Gross Profit
- Profit Margin
- Inventory Value
- Freight Cost
- Stock Turnover



## 4.3 SQL Processing Workflow

### Vendor Performance Analytics – ETL Workflow

End-to-End Data Processing Pipeline



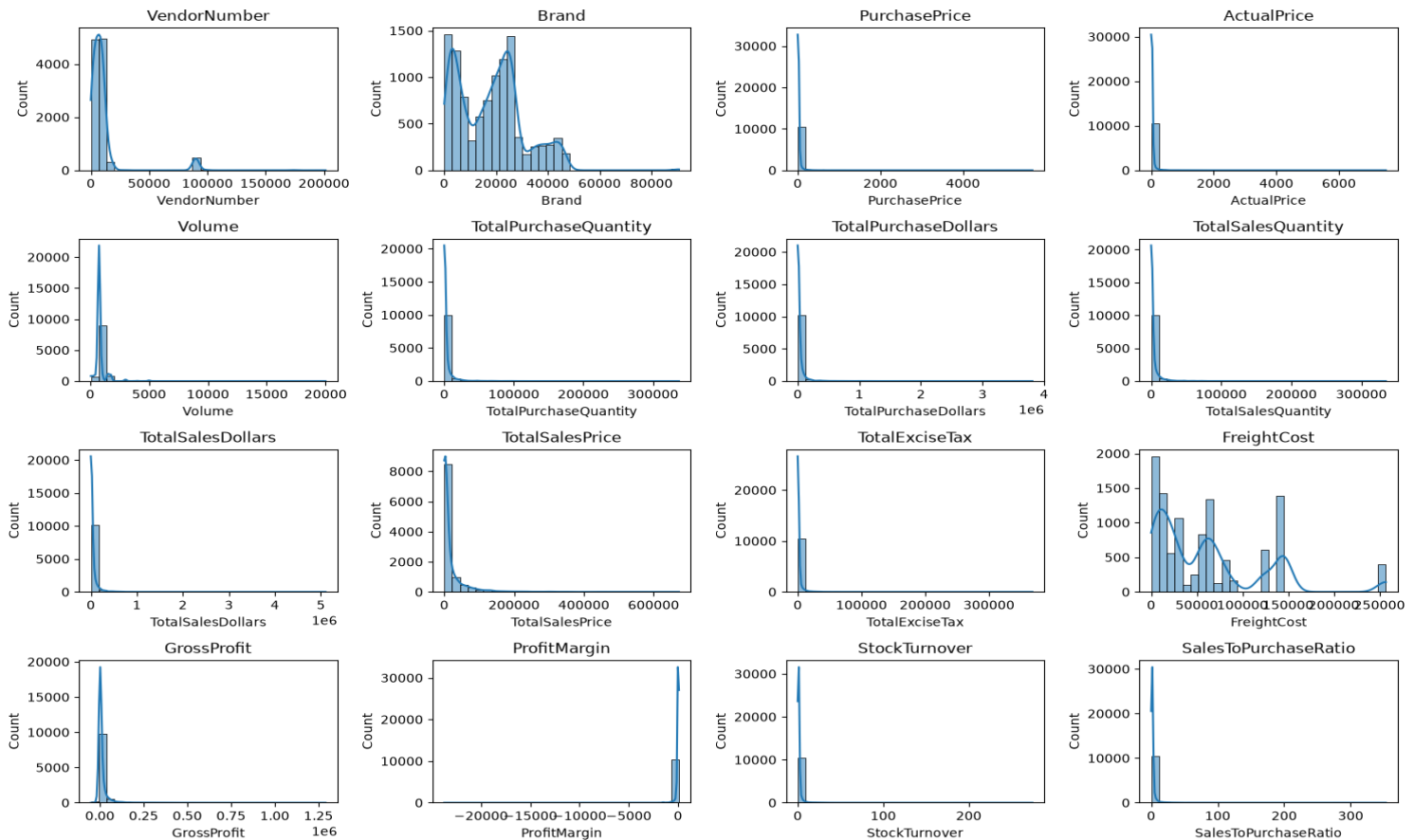
## 5.1 Introduction

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the vendor performance dataset and identify meaningful business patterns. Various statistical and visualization techniques were used to examine distributions, detect outliers, evaluate relationships between variables, and assess vendor performance. The insights obtained from EDA provide the basis for data-driven business recommendations.

## 5.2 Distribution of Numerical Variables

### Objective

This analysis examines the distribution of key numerical variables to understand the overall characteristics of vendor performance, procurement, sales, profitability, and inventory-related metrics. Distribution analysis helps identify data concentration, variability, and the presence of skewness that may influence business decisions.



**Figure 5.1:** Distribution plots illustrating the frequency distribution and kernel density estimation (KDE) of key numerical variables related to procurement, sales, profitability, freight cost, and inventory management.

## Observation

The distribution plots illustrate the spread and frequency of key numerical variables, including **Purchase Price, Total Purchase Dollars, Total Sales Dollars, Gross Profit, Freight Cost, Inventory Value, and Stock Turnover**. Most variables exhibit a positively skewed distribution, indicating that while the majority of vendors operate within moderate business volumes, a relatively small number of vendors contribute exceptionally high values.

Variables such as **Purchase Price, Sales Revenue, and Gross Profit** display long right tails, suggesting the presence of high-value transactions associated with a limited number of large vendors. Similarly, freight cost and inventory-related metrics show greater variability, reflecting differences in procurement volumes and inventory management practices across vendors.

## Business Interpretation

The positively skewed distributions indicate that business performance is concentrated among a relatively small group of high-performing vendors, while the majority contribute moderate transaction volumes. This pattern is common in procurement and sales environments, where a few strategic vendors account for a substantial portion of overall business activity.

Understanding these distributions enables management to distinguish between normal operational behaviour and exceptionally high-performing vendors. It also supports more effective procurement planning, vendor segmentation, and inventory optimization by identifying where business activity is concentrated.

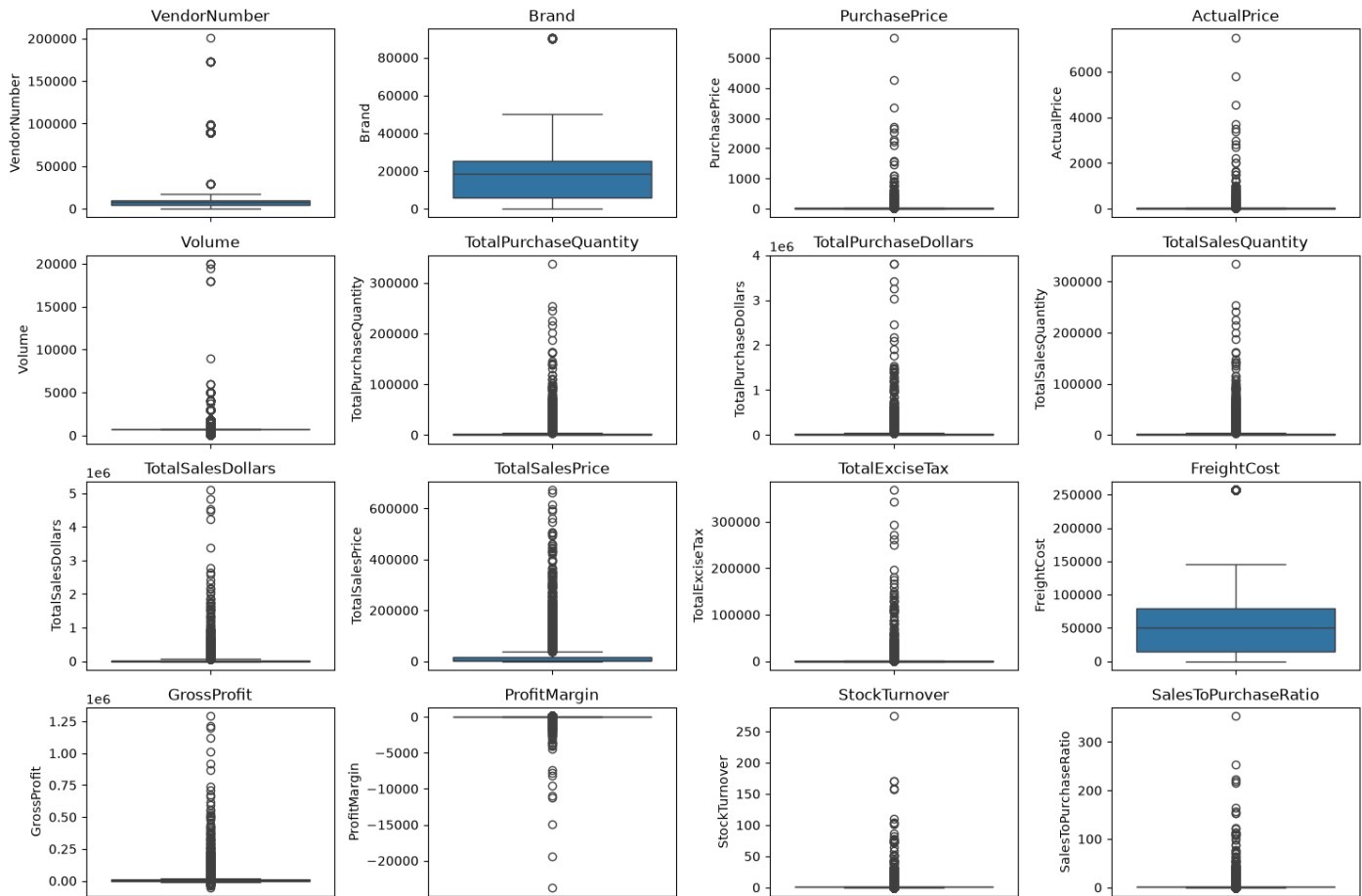
## Key Findings

- Most numerical variables exhibit a **positively skewed distribution**, indicating the presence of a limited number of high-value vendors.
  - Sales, purchase, and profit metrics are concentrated among relatively few vendors, while the majority operate within moderate business volumes.
  - Freight and inventory-related variables demonstrate higher variability, reflecting differences in procurement and inventory management practices.
  - Distribution analysis provides the foundation for identifying outliers and performing subsequent statistical analyses.
-

## 5.3 Outlier Detection

### Objective

This analysis identifies extreme values across key procurement, sales, profitability, and inventory-related variables. Detecting outliers helps distinguish exceptional business performance from normal operational patterns and supports more informed vendor evaluation.



**Figure 5.2:** Box plots illustrating the distribution of key procurement, sales, profitability, and inventory-related variables, highlighting vendors with exceptionally high business performance through statistical outliers.

### Observation

Box plots were used to examine the distribution of major business metrics, including Purchase Price, Total Purchase Dollars, Total Sales Dollars, Gross Profit, Freight Cost, Stock Turnover, and Sales-to-Purchase Ratio. Several outliers were observed across these variables, particularly in purchase costs, sales revenue, and gross profit, indicating the presence of vendors with exceptionally high transaction volumes and financial performance.

While most vendors fall within a relatively consistent operating range, a small number of vendors exhibit substantially higher values than the overall distribution. These extreme observations reflect the unequal contribution of vendors to procurement and sales activities.

### **Business Interpretation**

The identified outliers are not necessarily data anomalies or errors but instead represent strategically important vendors that contribute significantly to the organization's procurement and revenue. Such vendors often account for a disproportionate share of sales, purchases, and profits due to larger transaction volumes or stronger market demand.

Recognizing these high-performing vendors enables management to prioritize supplier relationships, negotiate favourable contracts, and ensure supply continuity. At the same time, organizations should monitor excessive dependence on these vendors, as concentration among a few suppliers may increase operational and supply chain risks.

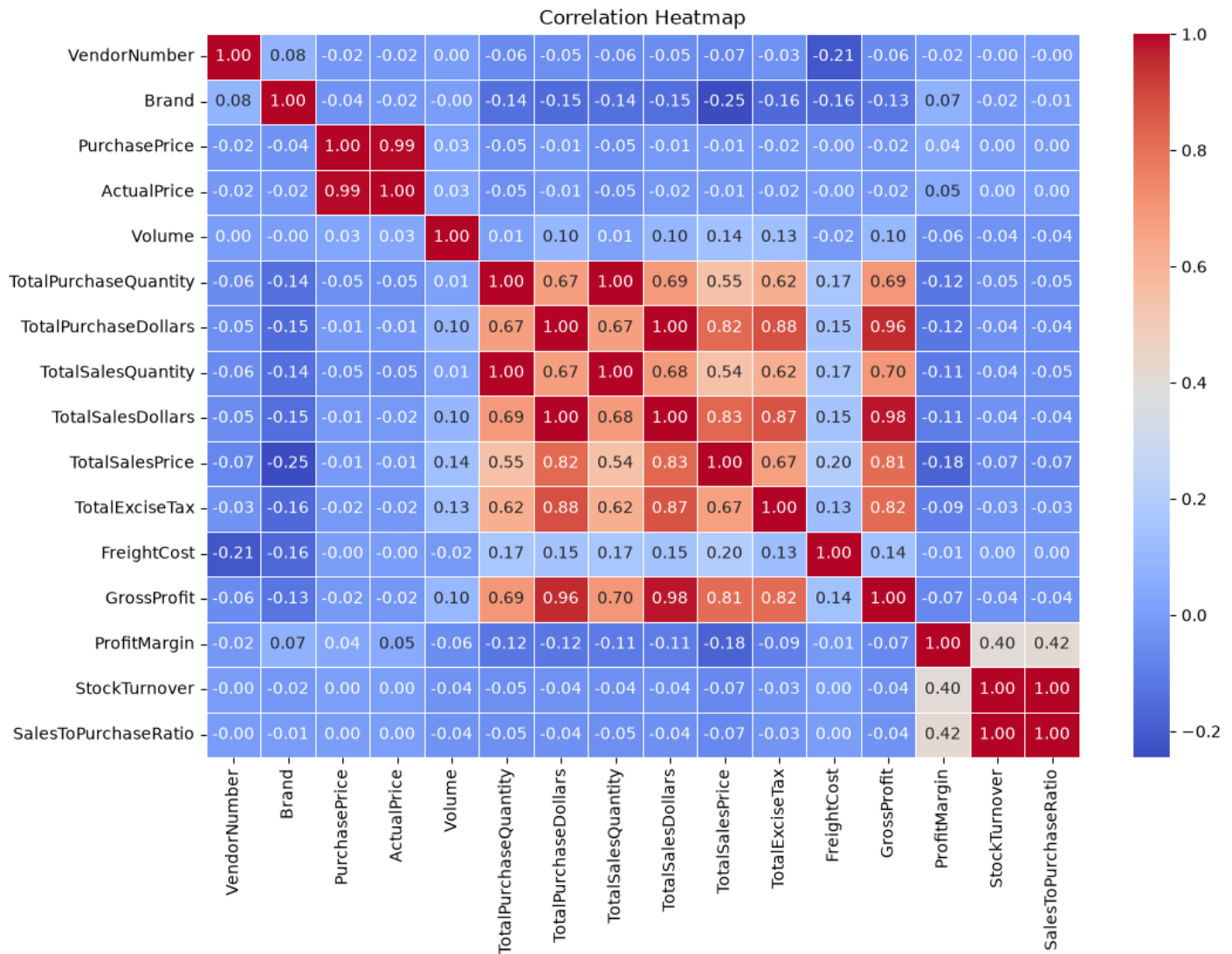
### **Key Findings**

- Multiple outliers were identified across purchase cost, sales revenue, and gross profit variables.
- The outliers primarily represent high-performing vendors rather than erroneous observations.
- Exceptional vendor performance contributes significantly to overall business revenue and profitability.
- Outlier analysis supports strategic vendor management, procurement planning, and risk assessment.

## 5.4 Correlation Analysis

### Objective

This analysis examines the relationships among key numerical variables associated with procurement, sales, profitability, inventory, and freight costs. Understanding these relationships helps identify the factors that most strongly influence vendor performance and supports more informed business decision-making.



**Figure 5.3:** Correlation heatmap illustrating the strength and direction of relationships among key procurement, sales, profitability, inventory, and freight-related variables.

### Observation

The correlation heatmap illustrates the strength and direction of relationships among major numerical variables. A strong positive correlation was observed between **Total Purchase Dollars** and **Total Sales Dollars**, indicating that vendors with higher procurement volumes generally generate higher sales revenue. **Gross Profit**

also exhibits a positive correlation with sales, suggesting that increased sales volume contributes to higher overall profitability.

Inventory-related variables, purchase quantity, and freight costs also demonstrate positive relationships with procurement activity, reflecting the operational impact of larger purchasing volumes. While many variables move together, the strength of their correlations varies, highlighting the complexity of vendor performance and procurement dynamics.

### **Business Interpretation**

The correlation analysis confirms that procurement volume is a key driver of sales performance, as vendors with higher purchase investments typically achieve greater revenue generation. However, strong sales do not always translate into proportionally higher profitability, emphasizing the need to evaluate both **revenue** and **profit margin** when assessing vendor performance.

These relationships provide valuable insights for procurement planning and vendor evaluation. Rather than relying solely on sales figures, organizations should adopt a balanced approach that considers purchasing costs, gross profit, inventory efficiency, and operational expenses to identify truly high-performing vendors.

### **Key Findings**

- A strong positive correlation exists between **purchase value** and **sales revenue**, indicating that higher procurement volumes generally lead to increased sales.
- **Gross profit** is positively associated with sales, although higher sales do not always guarantee higher profitability.
- Inventory and freight-related variables also exhibit positive relationships with procurement activity.
- Correlation analysis provides valuable insights for evaluating vendor performance across multiple financial and operational dimensions rather than relying on a single performance metric.

## 5.5 Top Vendors and Brands by Sales

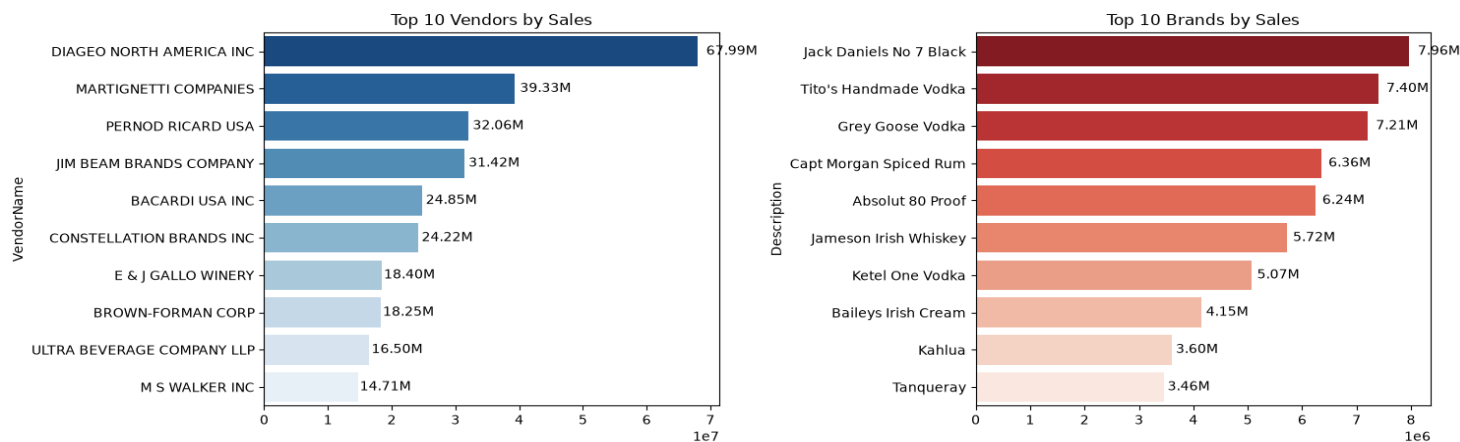


Figure 5.4: Top 10 vendors and top 10 brands ranked by total sales revenue.

### Observation

The analysis identifies the top-performing vendors and brands based on total sales revenue.

Among all vendors, **DIAGEO NORTH AMERICA INC** generated the highest sales, contributing approximately **67.99 million**, followed by **MARTIGNETTI COMPANIES (39.33 million)**, **PERNOD RICARD USA (32.06 million)**, and **JIM BEAM BRANDS COMPANY (31.42 million)**. The sales distribution shows a noticeable gap between the highest-performing vendor and the remaining vendors, indicating a strong concentration of revenue among a few suppliers.

Similarly, at the brand level, **Jack Daniels No. 7 Black** recorded the highest sales of approximately **7.96 million**, followed by **Tito's Handmade Vodka (7.40 million)**, **Grey Goose Vodka (7.21 million)**, and **Captain Morgan Spiced Rum (6.36 million)**. The top ten brands collectively represent a significant portion of total sales, demonstrating that customer demand is concentrated around a limited set of premium products.

### Business Interpretation

The results indicate that business revenue is highly dependent on a relatively small group of vendors and brands. Maintaining strong partnerships with these key vendors is essential for ensuring consistent product availability and maximizing revenue. Since a few premium brands contribute significantly to overall sales, inventory planning and procurement strategies should prioritize maintaining adequate stock levels for these products to avoid stock-outs.

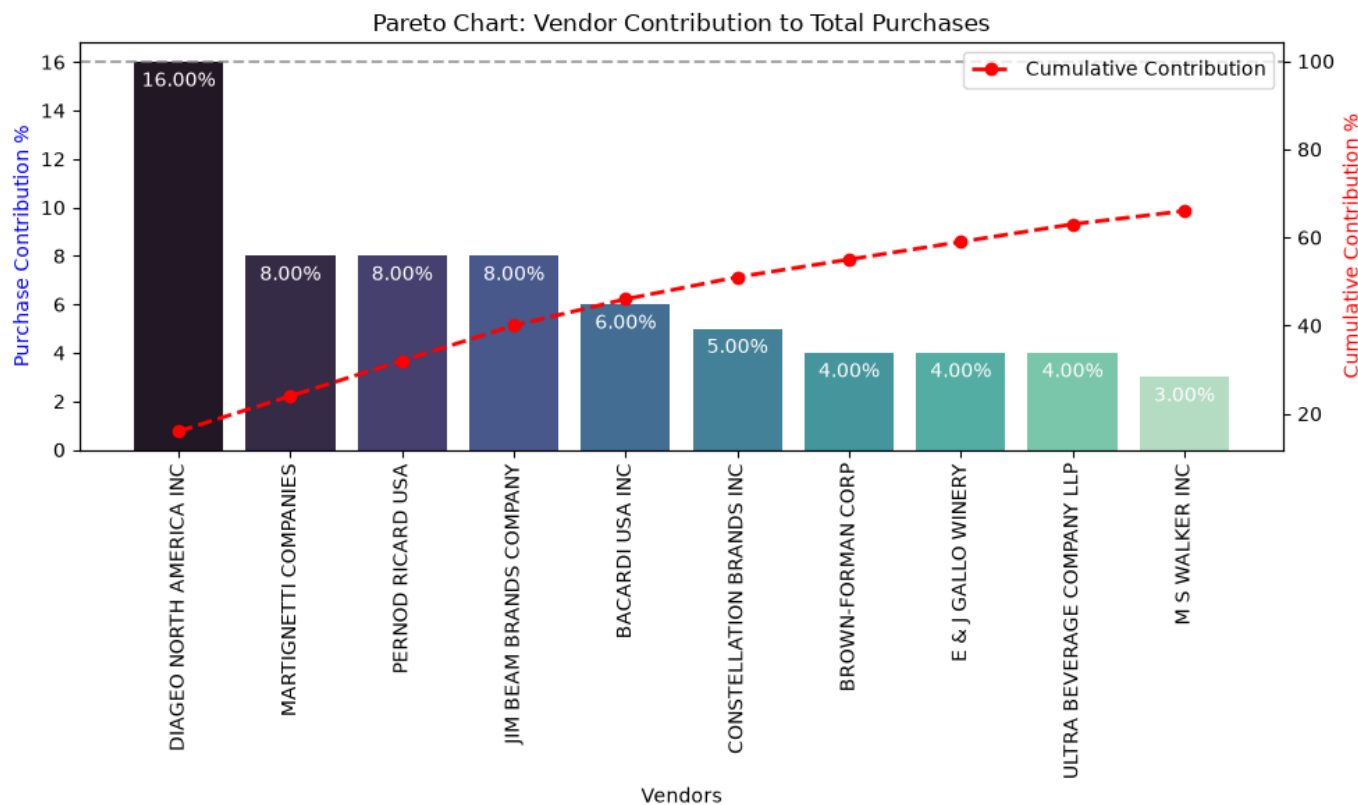
At the same time, the organization should evaluate opportunities to promote lower-performing brands and diversify its vendor base. Reducing dependence on a handful of vendors can improve supply chain resilience and minimize business risk associated with vendor-specific disruptions.



Key Findings

- **DIAGEO NORTH AMERICA INC** is the highest revenue-generating vendor with **67.99 million** in sales.
- The top four vendors contribute a substantially larger share of revenue than the remaining vendors.
- **Jack Daniels No. 7 Black** is the highest-selling brand with **7.96 million** in sales.
- Premium liquor brands dominate the top-selling product list.
- Sales are concentrated among a limited number of vendors and brands, highlighting the importance of strategic vendor management and inventory optimization.

5.6 Pareto Analysis of Vendor Contribution



**Figure 5.5:** Pareto analysis showing the percentage and cumulative contribution of the top vendors to total purchases.

## Observation

The Pareto chart illustrates the percentage contribution of the top vendors to the organization's total purchases along with the cumulative contribution curve.

Among all vendors, **DIAGEO NORTH AMERICA INC** contributes the highest share of purchases at **16%**, making it the single largest supplier. It is followed by **MARTIGNETTI COMPANIES**, **PERNOD RICARD USA**, and **JIM BEAM BRANDS COMPANY**, each contributing approximately **8%**. The remaining vendors individually contribute between **3% and 6%** of the total purchases.

The cumulative contribution curve demonstrates that purchase volume is concentrated among a relatively small number of vendors, while the remaining vendors account for progressively smaller portions of total procurement.

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## Business Interpretation

The Pareto analysis confirms that procurement activities are concentrated among a limited set of strategic vendors. Since a significant portion of purchases comes from only a few suppliers, maintaining strong relationships, negotiating favourable pricing, and ensuring consistent supply from these vendors are critical for operational efficiency.

At the same time, this concentration introduces supply chain risk. Heavy dependence on a small number of vendors may expose the organization to disruptions caused by pricing changes, delivery delays, or vendor-specific issues. Diversifying procurement across additional reliable vendors can help improve supply chain resilience while maintaining cost efficiency.

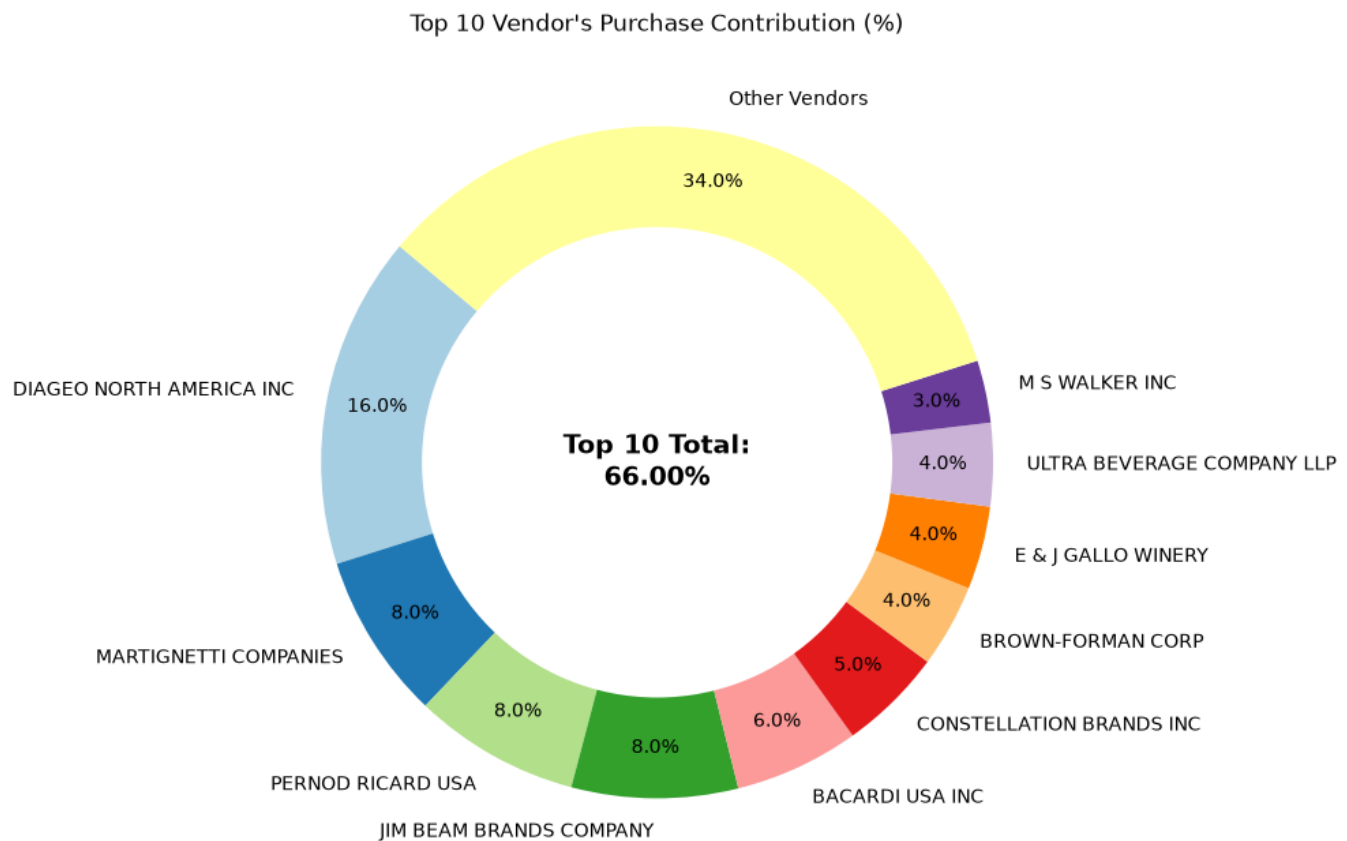
The Pareto principle also helps management identify high-priority vendors that should receive focused attention during vendor performance reviews, contract negotiations, and inventory planning.

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## Key Findings

- **DIAGEO NORTH AMERICA INC** alone contributes approximately **16%** of total purchases.
- The next three vendors each contribute around **8%**, indicating that procurement is concentrated among a small group of suppliers.
- The cumulative contribution curve highlights the unequal distribution of purchase volume across vendors.
- Strategic vendor management should prioritize high-contributing vendors while reducing excessive dependency on a limited supplier base.

5.7 Vendor Purchase Contribution Analysis



**Figure 5.6:** Percentage contribution of the top ten vendors to total purchases, highlighting procurement concentration across key suppliers.

Observation

The donut chart illustrates the percentage contribution of the top ten vendors to the organization's total purchase volume. Collectively, these vendors account for **66%** of total purchases, while the remaining vendors contribute the remaining **34%**.

Among all vendors, **DIAGEO NORTH AMERICA INC** contributes the largest individual share at **16%**. **MARTIGNETTI COMPANIES**, **PERNOD RICARD USA**, and **JIM BEAM BRANDS COMPANY** each contribute approximately **8%**, followed by **BACARDI USA INC** (**6%**) and **CONSTELLATION BRANDS INC** (**5%**). The remaining vendors individually contribute between **3% and 4%** of total purchases.

Business Interpretation

The analysis reveals that nearly **two-thirds of the organization's procurement is concentrated among just ten vendors**, highlighting their strategic importance within the supply chain. These vendors play a significant role in maintaining inventory availability and supporting business operations.

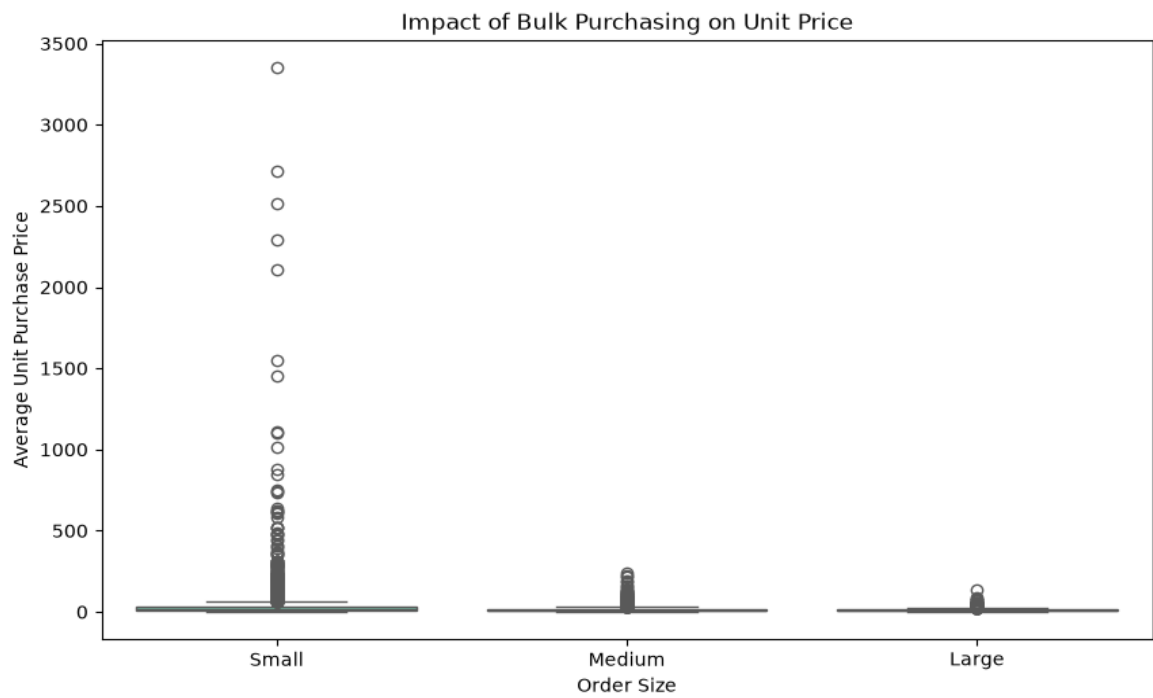
This level of concentration provides opportunities to strengthen long-term partnerships, negotiate better pricing, and improve procurement efficiency. However, it also emphasizes the need for supplier diversification to reduce dependency on a limited number of vendors and minimize operational risks associated with supply disruptions.

The visualization provides management with a clear understanding of vendor concentration and helps prioritize vendor relationship management based on procurement impact.

### Key Findings

- The **top 10 vendors contribute 66%** of the organization's total purchases.
- **DIAGEO NORTH AMERICA INC** is the largest contributor with a **16%** purchase share.
- Three additional vendors each contribute approximately **8%**, indicating a moderate concentration among leading suppliers.
- The remaining vendors collectively account for **34%** of purchases, demonstrating a long-tail distribution.
- Procurement decisions should focus on balancing strategic partnerships with supplier diversification to improve supply chain resilience.

### 5.8 Impact of Bulk Purchasing on Unit Price



**Figure 5.7:** Comparison of average unit purchase prices across different order-size categories, illustrating the impact of bulk purchasing on procurement costs.

## Observation

The box plot compares the average unit purchase price across three order-size categories: **Small, Medium, and Large**.

The analysis indicates that **small orders exhibit the highest variability in unit purchase prices**, with several extreme outliers reaching substantially higher values than the rest of the observations. Medium-sized orders show a narrower price distribution with fewer high-value outliers, while large orders demonstrate the **lowest median unit purchase price and the smallest variation** among the three categories.

Overall, the distribution suggests a decreasing trend in unit purchase price as order size increases.

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## Business Interpretation

The results indicate that **bulk purchasing is associated with lower average unit prices**, suggesting that vendors offer better pricing for larger order quantities. This supports the presence of volume-based purchasing benefits, where economies of scale help reduce procurement costs.

The high variability observed in small orders may be due to specialty products, emergency purchases, or low-volume items that are procured at premium prices. Whenever operationally feasible, consolidating smaller orders into medium or large purchase orders can improve cost efficiency and reduce overall procurement expenses.

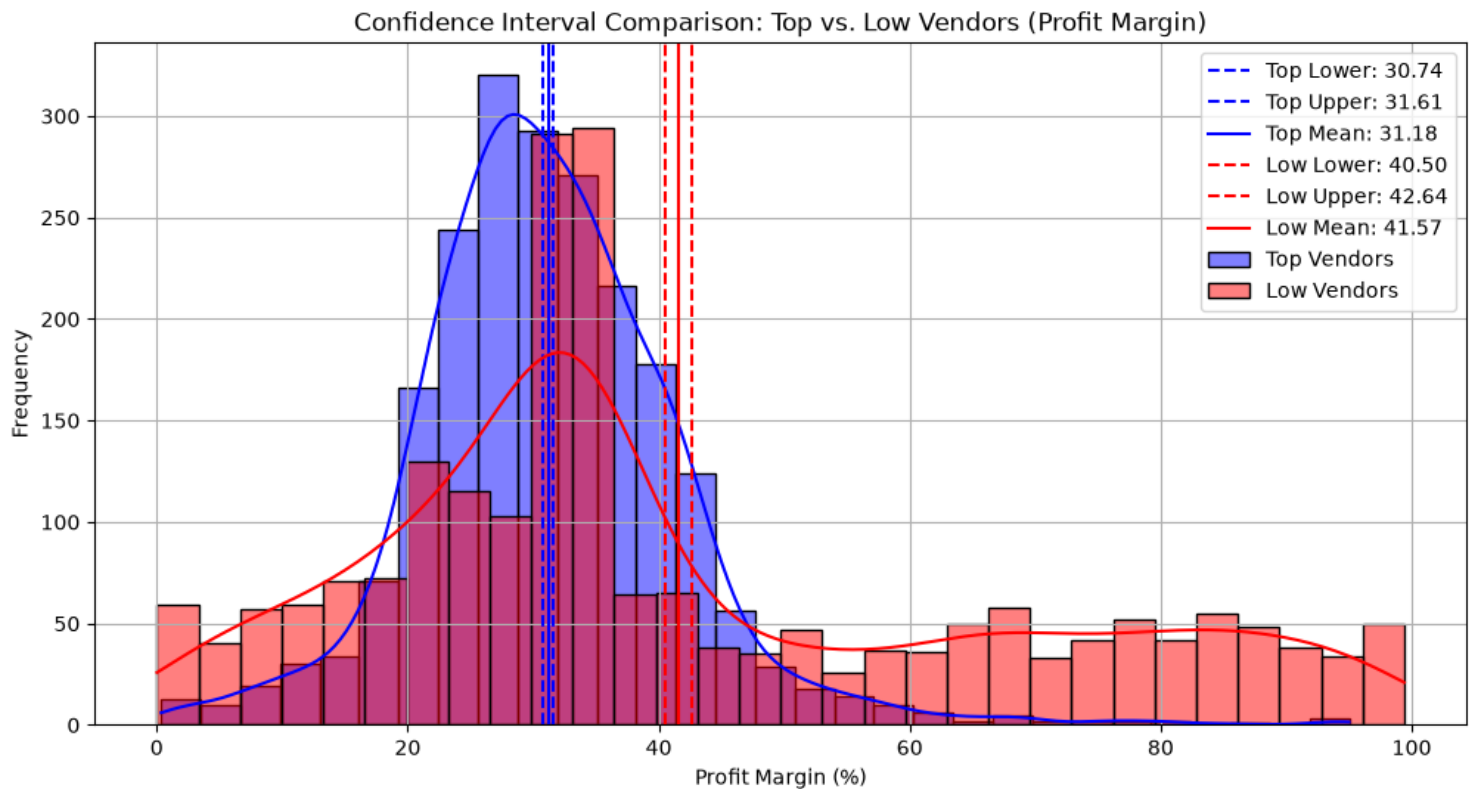
These findings can assist procurement managers in developing purchasing strategies that balance inventory requirements with cost optimization.

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## Key Findings

- Large orders have the **lowest average unit purchase prices**.
- Small orders show the **highest price variation** and contain numerous high-value outliers.
- Price variability decreases as order size increases.
- Bulk purchasing appears to provide measurable cost advantages through lower unit prices.
- Consolidating purchase orders may help reduce procurement costs while improving purchasing efficiency.

## 5.9 Confidence Interval Analysis of Profit Margin



**Figure 5.8:** Comparison of profit margin distributions and 95% confidence intervals for top-performing and low-performing vendors.

### Observation

The figure compares the distribution of profit margins between **top-performing vendors** and **low-performing vendors** using histograms, density curves, and **95% confidence intervals** for the mean profit margin.

The **top vendors** exhibit an average profit margin of approximately **31.18%**, with a narrow 95% confidence interval ranging from **30.74% to 31.61%**, indicating relatively consistent profitability across leading vendors.

In comparison, the **low-performing vendors** have a higher average profit margin of approximately **41.57%**, with a confidence interval between **40.50% and 42.64%**. However, their profit margins display substantially greater variability, as reflected by the wider spread of the distribution.

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### Business Interpretation

Although low-performing vendors exhibit a higher average profit margin, they contribute comparatively lower business volume. This suggests that products supplied by these vendors may carry higher margins but experience lower sales demand or limited market penetration.

Conversely, the top-performing vendors generate larger sales volumes while maintaining relatively lower yet more stable profit margins. This reflects a high-volume, moderate-margin business model that contributes more consistently to overall revenue.

The narrow confidence interval for top vendors indicates predictable profitability, making them more reliable for long-term procurement and revenue planning. On the other hand, the greater variability among low-performing vendors suggests higher uncertainty and may require further evaluation before expanding procurement from these suppliers.

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## Key Findings

- **Top vendors:** Mean profit margin  $\approx 31.18\%$  (95% CI: **30.74%–31.61%**).
- **Low-performing vendors:** Mean profit margin  $\approx 41.57\%$  (95% CI: **40.50%–42.64%**).
- Top vendors exhibit **more consistent and predictable profitability**.
- Low-performing vendors offer **higher margins** but with greater variability and lower business volume.
- Procurement strategies should balance **sales volume** and **profitability** rather than relying solely on margin percentages.

## 5.10 Inventory Turnover Analysis

### Objective

This analysis evaluates vendor-wise stock turnover to identify slow-moving inventory. Low stock turnover indicates that products remain in inventory for extended periods, leading to higher holding costs and inefficient utilization of working capital. Identifying such vendors helps improve inventory management and procurement planning.

### Observation

Stock turnover values were analysed across all vendors to identify those with turnover ratios below one. Vendors exhibiting low turnover indicate that purchased inventory is not being converted into sales efficiently, resulting in excess stock accumulation.

### Business Interpretation

Low inventory turnover suggests overstocking, weak customer demand, or inefficient procurement planning. Maintaining excess inventory increases storage costs, blocks working capital, and raises the risk of product obsolescence. Monitoring these vendors enables the organization to optimize purchase quantities and improve inventory efficiency.

### Key Findings

- Vendors with stock turnover below one requires closer inventory monitoring.

- Slow-moving inventory increases inventory carrying costs.
  - Procurement decisions should be aligned with actual sales demand to improve turnover ratios.
- 

## **5.11 Unsold Inventory Value Analysis**

### **Objective**

This analysis estimates the monetary value of inventory that remains unsold after procurement. Measuring vendor-wise unsold inventory helps identify suppliers contributing the most to capital locked in inventory and supports more effective inventory optimization strategies.

### **Observation**

Unsold inventory value was calculated by comparing purchased quantities with sold quantities and multiplying the remaining inventory by the purchase price. The vendor-level aggregation highlights suppliers responsible for the largest amount of capital tied up in unsold inventory.

### **Business Interpretation**

High unsold inventory indicates inefficient inventory utilization and excessive capital investment in products that have not yet generated revenue. Vendors with consistently high unsold inventory should be reviewed to optimize purchasing frequency, improve demand forecasting, and reduce inventory carrying costs.

### **Key Findings**

- Significant working capital remains locked in unsold inventory across selected vendors.
- Vendor-wise inventory monitoring can reduce unnecessary procurement.
- Better demand forecasting and optimized reorder quantities can improve cash flow and inventory efficiency.



Based on the analytical findings, the following recommendations are proposed to improve vendor performance and overall procurement efficiency:

1. Diversify procurement across additional vendors to reduce dependency on a limited number of high-contributing suppliers.
2. Increase bulk purchasing for products demonstrating consistent demand to benefit from lower unit purchase costs.
3. Implement targeted promotional campaigns for brands exhibiting high profit margins but relatively low sales performance.
4. Continuously monitor vendors with low inventory turnover and adjust procurement quantities based on historical sales trends.
5. Regularly review vendors contributing significantly to unsold inventory and optimize replenishment schedules to reduce capital lock-in.
6. Establish vendor performance dashboards that track sales, profitability, inventory turnover, procurement contribution, and gross profit to support data-driven decision making.
7. Periodically evaluate vendor profitability using statistical methods such as confidence interval analysis to ensure procurement strategies remain aligned with business objectives.

This project demonstrates a comprehensive vendor performance analysis using Python, SQL, SQLite, and Pandas by integrating multiple procurement, sales, and inventory datasets into a unified analytical framework. Through ETL processing, SQL-based aggregation, exploratory data analysis, and statistical evaluation, the study identified key factors influencing vendor performance, procurement efficiency, inventory utilization, and profitability.

The analysis revealed opportunities to optimize purchasing through bulk procurement, reduce dependency on a limited number of vendors, improve inventory turnover, minimize capital locked in unsold inventory, and promote high-margin products with lower sales performance. Statistical techniques further validated differences in vendor performance, strengthening the reliability of the business insights.

Overall, this project demonstrates how data analytics can transform large-scale operational data into actionable business intelligence, enabling organizations to make informed procurement decisions, improve inventory management, and enhance overall operational efficiency.

## **Project Repository**

GitHub : <https://github.com/Kush91-bit/Vendor-Performance-Analysis>